



TOBB University of Economics and Technology

Department of Computer Engineering

BIL 481 - Software Engineering

Fall 2025 Semester

Universal Software-Driven Constraint Tracking Form

NOTLA

University Lecture Notes Sharing and Evaluation Platform

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UNIVERSAL SOFTWARE-DRIVEN CONSTRAINT TRACKING FORM

Constraint ID	Category	Constraint Description	Design Rationale	Owner	Impacted Requirements	Domain	Impacted Components	Associated Standards	Verification Method	Risk Impact	Status
SWC-01	Performance	The course list and course detail API endpoints must respond within ≤ 1 second for 95% of all requests. The React frontend must render each page within ≤ 2 seconds.	Required to meet user experience expectations and to comply with SRS Performance Requirements (PRF1–PRF2). Ensures stable operation on low-cost hosting environments.	Backend Lead (Yunus Emre Özçelik)	PRF1, PRF2	Web, Cloud	Django REST API, React UI, Database Query Layer	IEEE 29148; ISO/IEC 25010 – Performance Efficiency	Load testing, response-time measurement	High	Active
SWC-02	Security	User passwords must never be stored in plaintext. All passwords must be stored using bcrypt or PBKDF2 hashing with salt. No log or system component may contain plaintext passwords or password-derived information.	Required for compliance with GDPR/KVKK and SRS Security Requirements (SEC1–SEC3). Prevents credential exposure in the event of a data breach.	Security Architect (Abdullah Arda Gündoğdu)	SEC1, SEC3	Web, Cloud	Authentication Module, User Model, Logging Layer	GDPR; KVKK; ISO/IEC 27001	Static code analysis, penetration testing	High	Active

SWC-03	Resource Constraint	The maximum allowed size for an uploaded file is 25 MB. Total storage consumption must remain within the limits of the free-tier hosting services used by the project.	The SRS defines a strict financial constraint (SYS6 – 0 TL budget). Limiting file sizes ensures the application stays within free-tier storage limits.	DevOps / Database Administrator	FR4, SYS6	Web	File Upload Service, Storage Layer	IEEE 29148 – Resource Constraints	Upload-size validation, multi-file stress testing	Medium	Active
SWC-04	Compatibility / Platform	The system must operate only on modern desktop browsers (Chrome, Firefox, Edge — last two versions). Mobile application support and full mobile optimization are excluded from this development phase.	The SRS explicitly states (SYS1) that the system is web-only and mobile support is outside the scope of the current release.	Frontend Lead (Ömer Talha Akbulut)	SYS1, UI1	Web	React UI, Router, Authentication Screens	ISO/IEC 25010 – Compatibility	Cross-browser smoke testing	Medium	Active

SWC-05	Performance & Reliability	The system must maintain acceptable performance levels (as defined in SWC-01) while supporting at least 100 concurrent users.	This concurrency requirement is explicitly stated in SRS Performance Requirements (PRF3). Ensures platform reliability during peak academic periods.	Backend Lead (Yunus Emre Özçelik)	PRF3, PRF4	Web, Cloud	API Gateway, Session Manager, Database Connection Pool	IEEE 29148 – Performance Constraints	Concurrency load testing	High	Active
SWC-06	Security & Privacy	User data (email, IP address, logs) must be stored in full compliance with KVKK/GDPR. Email addresses must be masked in logs. All personal data must be retained only for the minimum legally necessary duration.	Required by SRS Legal Requirements (LEG1–LEG2). Minimizing personally identifiable information (PII) in system logs reduces legal and security risks.	Security Architect / System Administrator (İsmail Emre Yıldız)	SEC2, SEC3, LEG1	Web, Backend	Logging System, User Module, Admin Tools	KVKK 6698; GDPR	Log inspection, privacy compliance checklist	High	Proposed

SWC-07	Human Factors / UX	Core user actions (course search, viewing notes, downloading notes) must be achievable within 3–4 clicks. The user interface must be delivered in Turkish only for this development phase.	Based on SRS UI and Usability Requirements (FR10, UI2, UX1). Simplifies onboarding for the target user population (Turkish university students).	UX Lead (Ömer Talha Akbulut)	FR10, UI2, UX1	Web	Navigation Bar, Search Component, Course Detail Interface	ISO/IEC 25010 – Usability	Scenario-based usability testing	Medium	Active
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